

HIMANSHU NAKRANI

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PROFESSIONAL SUMMARY

AI Engineer with over 2 years of experience architecting and deploying production-grade Generative AI and NLP applications, agentic AI systems, and RAG pipelines. Proven expertise in fine-tuning large language models, developing sophisticated Text-to-SQL systems, and engineering scalable AI inference systems that drive significant performance and efficiency improvements for enterprise AI solutions.

TECHNICAL SKILLS

Generative AI & LLMs: Large Language Models, RAG (Retrieval-Augmented Generation), Agentic AI, Multi-agent Systems, AI Guardrails, Responsible AI, Text-to-SQL, Fine-tuning (LoRA/PEFT), TRL, Prompt Engineering, Embeddings, Model Optimization, LangChain, LangGraph, Hugging Face, OpenAI API, Azure OpenAI, Google ADK
Backend Development: FastAPI, ASP.NET Core, REST APIs, SQLAlchemy, Microservices, API Design, OOP
Databases: SQL, Vector Databases (pgvector), Semantic Search, Query Optimization
Machine Learning: Deep Learning, NLP, Transformers, PyTorch, JAX/MaxText, Data Pipelines, Model Deployment, Data Augmentation
Cloud Platforms: Azure, AWS, Google Cloud TPU, Oracle Cloud Infrastructure (OCI), Databricks, Azure OpenAI Studio
Programming Languages: Python, Java, SQL, C/C++, C#, Shell/Perl Scripting
Developer Tools: Git, Docker, Harness, CI/CD, MLOps, LLMops, Model Observability, Testing, Evaluation Frameworks

PROFESSIONAL EXPERIENCE

State Street Corporation

Hyderabad, India

Senior Associate - AI Software Developer

July 2023 – Present

- Architected and deployed "**Alpha Copilot**", an enterprise LLM-powered Text-to-SQL chatbot enabling natural language queries over structured financial data, serving 100+ internal users
- Engineered scalable AI inference infrastructure using **FastAPI** and **SQLAlchemy**, implementing advanced caching and **latency optimization** techniques that reduced average response time by **75% (from 25-30s to 6-8s)**
- Developed production-ready **RAG system** for extracting actionable insights from fund prospectuses, engineering an optimized document pipeline with intelligent chunking, semantic embeddings, and vector database integration achieving sub-second retrieval times
- Integrated LLM-based data visualization capabilities, automatically generating interactive charts and summaries from query results, improving data accessibility and decision-making speed
- Led backend development for **WealthAI application**, building real-time portfolio narrative generation system that processes optimizer outputs and delivers personalized investment insights at scale
- Fine-tuned **GPT-4o mini**, **GPT-4.1 mini** (Azure OpenAI) and **Llama 3.1-8B** (Databricks) using supervised fine-tuning and PEFT/LoRA, managing end-to-end LLMops pipeline including data augmentation, training, and inference optimization
- Conducted research on advanced Text-to-SQL generation techniques, exploring schema-linking methods and few-shot prompting strategies to improve SQL accuracy by 25%
- Spearheaded development of "**Agent Forge**", an enterprise **agentic AI** platform centralizing agent registration, discovery, and governance — actively used by **5+ business teams** with **50+ registered agent workflows**, providing a unified inventory of metadata, capabilities, and performance indicators across **multi-agent systems**
- Designed and implemented OTEL-compliant **model observability**, advanced **AI guardrails** with input/output validation and content filtering, and automated evaluation pipelines for **Responsible AI** compliance across all registered agents; building a no-code agentic workflow builder for visually designing and deploying custom AI agents
- Achieved **95%+ unit test coverage** and streamlined deployment workflows by architecting automated **CI/CD pipelines** with **Harness** and **Docker**, significantly improving system reliability and reducing production incidents
- Collaborated with cross-functional teams in an Agile environment, partnering with business analysts and compliance officers to align deliverables with regulatory requirements and business objectives

State Street Corporation

Hyderabad, India

Software Development Intern

January 2023 – June 2023

- Led migration of legacy ideation platform from .NET Core to modern **FastAPI + ReactJS** stack, improving application performance by 40% and developer productivity
- Designed and implemented RESTful APIs with SQLite backend, ensuring seamless frontend integration and comprehensive test coverage

PUBLICATIONS & RESEARCH

GoT4SQL-DA: A Graph-of-Thoughts Framework for Scalable Text-to-SQL Data Augmentation	FLLM 2025
GRAFT: Graph-of-Thoughts based Reasoning and Augmentation for Finetuning Text-to-SQL	CSCI 2025

EDUCATION

Nirma University	Ahmedabad, India
Bachelor of Technology in Electronics and Communication Engineering	June 2019 – July 2023

KEY PROJECTS

TinyMathReason-1B: Mathematical Reasoning LLM JAX/MaxText, PyTorch, TRL, Cloud TPU, AMD MI300X, GCS, Python GitHub - Pretrained a 1.12B parameter Llama-style decoder-only transformer from scratch on a Google Cloud TPU v4-32 cluster using MaxText/JAX, processing a 300B token math-heavy dataset with a throughput of 66 TFLOP/s/device - Engineered a distributed data pipeline across parallelized cloud instances, processing 4 large-scale datasets (FineWeb-Edu, OpenWebMath, Proof-Pile-2, Stack-Edu) with MinHash deduplication and sequence packing into optimized binary shards on GCS - Built custom JAX/Orbax to Hugging Face checkpoint conversion utilities and implemented a two-stage post-training curriculum (SFT, DPO/GRPO) using TRL on an AMD MI300X GPU with custom tokenizer extensions for chain-of-thought reasoning
CogniDoc: RAG Document Q&A Platform FastAPI, Supabase pgvector, OpenAI, Gemini, Vercel Live - Developed a production-ready Retrieval-Augmented Generation (RAG) pipeline encompassing document ingestion, intelligent chunking, and context-aware semantic retrieval utilizing Supabase pgvector - Engineered a scalable FastAPI backend with multi-model LLM routing (OpenAI, Gemini) to optimize response latency and relevance, empowering real-time conversational memory in a Vercel-deployed application

CERTIFICATIONS

Oracle Cloud Infrastructure Certified Generative AI Professional View Certificate
Generative AI Fundamentals (Databricks) View Certificate
Amazon Machine Learning Summer School 2022 View Certificate
AWS Machine Learning Foundations (Udacity) View Certificate